

robot. A major challenge is ensuring the software is highly responsive. The robot operates at hundreds of control cycles per second, often at 100 or even 500Hz, meaning the system performs 500 calculations every second. At this speed, issues can arise if processing is too slow or if communication between systems is delayed. The software must also be optimised for low-power computing, since putting the most powerful computer on the robot would consume too much energy,” explains Aditya.

On talking about the revenue, Aditya says, “Including Cobot C1 and Svan M2, we now have 12 customers. Our total revenue so far has crossed 10 million rupees.”

Most assembly and system integration work occurs in Kanpur, while manufacturing is outsourced to select partners in Bengaluru. The startup provides these partners with drawings and technical guidance, and over iterations with the last four or five robots, the manufacturers have become well-versed in building these components. These partners are now long-term collaborators for contract manufacturing.

One of the biggest challenges the startup faces is talent. Funding is also a major factor, and the company has struggled to secure it, though some agreements are now close to finalisation. Talent is harder to find. Most of the skilled workforce has been developed in-house,



Aditya Pratap Singh Rajawat, xTerra Robotics

with employees hired and trained to deliver on complex robotics projects. There are very few professionals already experienced with these types of robots, so xTerra also works with research organisations such as IITs and NITs, supplying robots and training students while they are still doing B.Tech. Some students even interact directly with the team. This approach helps create a pipeline of trained talent for future hires, though the shortage remains significant.

“We have been supported by incubators. Our company was incubated at IIT Kanpur, and we have received support through programmes such as the Social Innovation Lab, which provides government grants for specific projects. We have also been supported by the I-Hub Foundation for Cobotics at IIT

Delhi, a robotics-focused incubator, which provided a convertible note to support the production and development of Svan M2. In addition, the ITEL Foundation gave us a grant to support product development. So far, we have not raised any external funding outside of these incubators or government programmes,” says Aditya.

On discussion about how the ecosystem is helping them, Aditya says, “Our reseller network is growing. Right now, we have resellers in Kanpur and Gujarat who help us reach government and private customers. We haven’t scaled to many channel partners across India yet, but the ecosystem is growing, particularly driven by the drone industry. These partners and integrators can help bring our robots to industrial customers. We also see robotics-as-a-service becoming common, similar to drones-as-a-service. The growing Indian drone ecosystem is helping us reach the right customers at the right time.”

Currently, the company’s focus is primarily on R&D and building industrial-grade robots. Current production is close to five robots per quarter. Over the next year, the goal is to reach 50 robots per quarter for one product line. Additional product lines are in development, and the long-term target is to scale to 1000 robots per quarter by establishing a dedicated factory.

—Nidhi Agarwal

3 MEASURING IN MILLIMETRES WHILE TRAVELLING AT 100KM/HR

An Indian deep-tech startup builds hardware-led 3D vision systems that measure roads and railway tracks with millimetre accuracy, delivering inspection-grade data beyond the reach of software-only AI.

C3D Vision Systems is a Mumbai-based deep-tech startup operating at the intersection of electronics, optics, AI, and infrastructure engineering. They design and build high-precision,

vehicle-mounted 3D vision systems that digitally capture, measure, and analyse linear assets such as roads, highways, rail tracks, runways, bridges, and related infrastructure.

They grew out of first-hand

experience on infrastructure projects, where maintenance decisions depended on manual surveys, visual inspections, or basic camera tools that could detect defects but not measure them accurately.